ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

DISTANCE LEARNING PROGRAMME
(Academic Session : 2024 - 2025)

JEE (Main)
TEST # 07
29-09-2024

JEE(Main+Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Review (Unit Test # 01, 02, 03, 04 & 05)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	A	C	A	A	D	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	C	A	B	C	C	D	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2	1	3	2	3	2	1	9	4	5

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	C	A	B	A	C	A	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	A	A	B	B	C	D	D	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	45	5	5	4	2	2	2	0	4	2

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	C	C	B	B	A	C	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	C	C	A	B	D	B	A	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1	5	1	2	1	8	2	2	6	1

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (C)

$$\text{Pressure} = \left[\frac{\text{Force}}{\text{Area}}\right] = \left[\frac{\text{Force} \times \text{m}}{\text{Area} \times \text{m}}\right] = \left[\frac{\text{Energy}}{\text{Volume}}\right]$$

2. Ans (B)

$$\cos\beta = \frac{A_y}{A} = \frac{-1}{\sqrt{(1)^2 + (-1)^2 + \left(\sqrt{2}\right)^2}} = -\frac{1}{2}$$

$\therefore \beta = 120°$

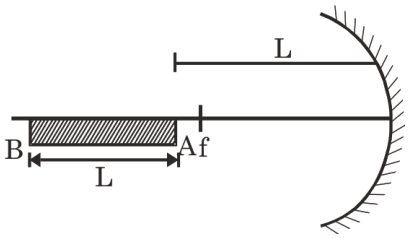
3. Ans (C)

$\mu_1 = \mu_3$ since there is no bending at first surface.

$\mu_3 < \mu_2$ since the ray bends towards normal as it passes from μ_3 to μ_2 medium.

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4. Ans (A)



m is magnification of point A distance of image from pole = $m_1 x$

$$\left[m_1 = \frac{f}{L-f}\right] = \frac{fL}{L-f}$$

Distance of image of point B from poll

$= m_2 \times 2\lambda$

$$\left[m_2 = \frac{f}{2L-f}\right] = \frac{2Lf}{2L-f}$$

$$\text{image length} = -\frac{2Lf}{2L-f} + \frac{Lf}{L-f}$$

$$= Lf\left[\frac{-2(L-f)+(2L-f)}{(2L-f)(L-f)}\right] = \frac{Lf \times f}{(2L-f)(L-f)}$$

5. Ans (C)

$$\tan 37 = \frac{2H}{R} = 2\left[\frac{u_y^2}{2g}\right]\left[\frac{g}{2u_x u_y}\right]$$

$$2\tan 37 = \frac{u_y}{u_x} = \tan\theta$$

$$2 \times \frac{3}{4} = \tan\theta$$

6. Ans (A)

$$\frac{da}{dt} = a$$

$a = 2t$

$v = t^2$

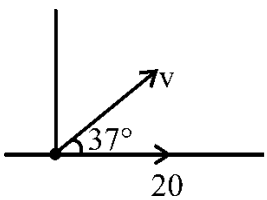
$$x = \frac{t^3}{3}$$

Distance travelled in 3rd sec

$$= x_3 - x_2 = \frac{27}{3} - \frac{8}{3} = \frac{19}{3}$$

7. Ans (A)

x component of velocity of both the ships are same.



$v\cos 37 = 20$

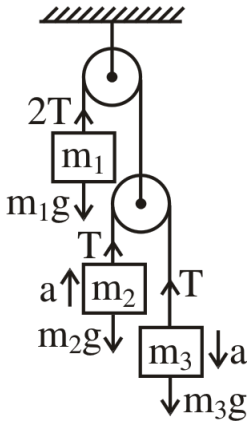
$$v \times \frac{4}{5} = 20$$

$v = 25$

8. Ans (D)

$$\text{flux} = \frac{1}{2}\left(\frac{q}{24\varepsilon_0}\right)$$

9. Ans (D)



$$T = \frac{2m_2 m_3 g}{m_2 + m_3} \quad ....(i)$$

$2T = m_1 g$ ....(ii)

10. Ans (B)

Applying WET

$$\frac{mg}{2}x - \int_0^x (\mu_0 x)mgdx = 0$$

We get $x = \frac{1}{\mu_0}$

11. Ans (A)

$$V_A = \frac{-GM}{3R};\ V_B = \frac{-GM}{2R^3}\left(3R^2 - \frac{R^2}{4}\right)$$

$$\frac{-11GM}{8R}$$

By conservation of energy

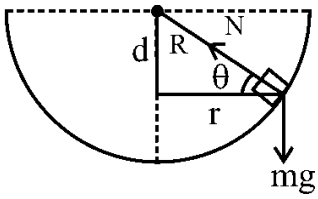
$$mV_A = mV_B + \frac{1}{2}mv^2$$

$$\Rightarrow \frac{-GMm}{3R} = \frac{-11GMm}{8R} + \frac{1}{2}mv^2$$

$$\Rightarrow v = \frac{5}{2}\sqrt{\frac{GM}{3R}}$$

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12. Ans (B)



Centripetal

Force = mass × centripetal acceleration

$N\cos\theta = m\frac{V^2}{r}$ ....(i)

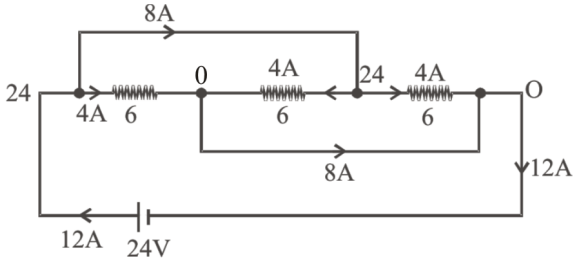
$N\sin\theta = mg$ ....(ii)

(ii)/(i), $\tan\theta = \frac{rg}{V^2}$

$\Rightarrow V = \sqrt{\frac{rg}{\tan\theta}}$

$V = \sqrt{\frac{g(R^2-d^2)}{d}}$ $\left[\begin{array}{c} r=\sqrt{R^2-d^2} \\ \tan\theta = d/r \end{array}\right]$

13. Ans (C)



$R_{eq} = 2\Omega$

[All 3 resistances are in parallel]

So total current = $\frac{24}{2}$ = 12 Amp.

14. Ans (A)

$R = \frac{\rho\ell}{A} = \left(\frac{m}{ne^2\tau}\right)\frac{\ell}{A}$

15. Ans (B)

$F_x = -\frac{\partial U}{\partial x} = \sin(x+y)$

$F_y = -\frac{\partial v}{\partial y} = \sin(x+y)$

$F_x = \sin(x+y)]_{(0,\,\pi/4)} = \frac{1}{\sqrt{2}}$

$F_y = \sin(x+y)]_{(0,\,\pi/4)} = \frac{1}{\sqrt{2}}$

$\therefore = \frac{1}{\sqrt{2}}\left[\hat{i}+\hat{j}\right]$

16. Ans (C)

$P = Fv = v(5-v) \Rightarrow F = 5-v$

$m\frac{dv}{dt} = 5 - v$

$\int_0^v \frac{m}{5-v}dv = \int_0^2 dt$

$[-m\ell n(5-v)]_0^v = 2$

$\Rightarrow -2\ell n\left(\frac{5-v}{5}\right) = 2 \Rightarrow 1-\frac{v}{5} = e^{-1}$

$v = 5\left(1-e^{-1}\right)$

Now $W = \Delta KE$

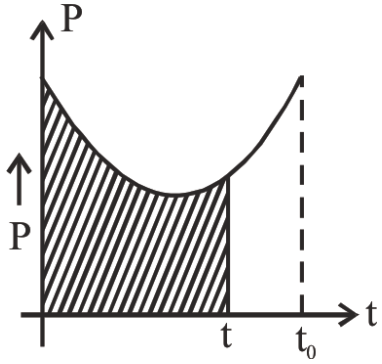
$= \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}\times 2\times 25\left(1-e^{-1}\right)^2$

$= 25\left(\frac{e-1}{e}\right)^2$

17. Ans (C)

The work done by force from time t = 0 to t = t sec. is given by shaded area in graph below.

Hence as t increases, this area increases.



∴ Work done by force keeps on increasing.

18. Ans (D)

Here $R_{eq} = R' = \frac{7R}{4}$

Now $i = \frac{q_0}{R'C}e^{-\frac{t}{R'C}}$

$\therefore \frac{1}{4}\frac{q_0}{R'C} = \frac{q_0}{R'C}e^{-\frac{t}{R'C}}$

or $\ell n\,4 = \frac{t}{R'C}$

$\therefore t = \frac{7RC}{4}\ell n4 = \frac{7RC}{2}\ell n2$

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19. Ans (A)

$C = \varepsilon_0 A/d$

$$\frac{1}{C'} = \frac{1}{C} + \frac{4}{C} + \frac{3}{C} = \frac{8}{C} \Rightarrow C' = \frac{C}{8}$$

$$C_{eq} = \frac{C}{2} + \frac{C}{8} = \frac{4C + C}{8} = \frac{5}{8}C$$

20. Ans (B)

$\vec{a}_A = 6ti - 8t^5 j$

at $t = 1$ sec ; $\vec{a}_A = 6i - 8j$

$$a_1 = 6t\hat{i} - \frac{4}{3} \times 6(t)^5\hat{j} = 6\hat{i} - 8\hat{j}$$

$\therefore$ Pseudo force on B w.r.t. A $\Big\} = -m_B \times \vec{a}_A$

$= -50\,[6i - 8j]$

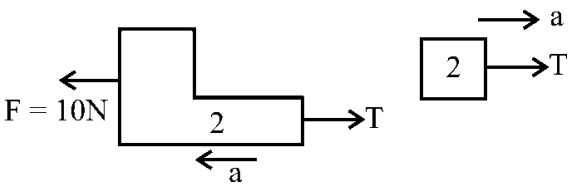
$= 50\,[8j - 6i]$

$= 100\,[4j - 3i]$

PART-1 : PHYSICS

SECTION-II

1. Ans (2)



$10 - T = 2a$ ... (i)

$T = 2a$ ... (ii)

From equation (i) and (ii)

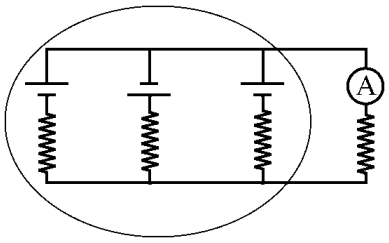
$$a = \frac{10}{4} = \frac{5}{2}$$

$a_{relative} = 5$

$$10 = \frac{1}{2}(5)\,t^2$$

$t = 2$

2. Ans (1)



Find equivalent

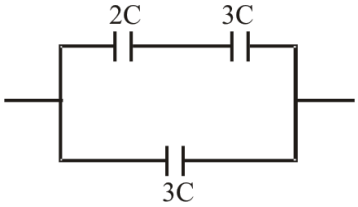
$$E = \frac{\frac{10}{2} - \frac{36}{3} + \frac{12}{3}}{\left(\frac{1}{2} + \frac{1}{6} + \frac{1}{3}\right)} = 3\ \text{Volt}$$

$$\Rightarrow i = \frac{3}{2+1}$$

$$r = \left(\frac{1}{2} + \frac{1}{6} + \frac{1}{3}\right)^{-1} = 1\Omega$$

3. Ans (3)

$$C_{eq} = \frac{21}{5} \times \frac{\epsilon_0 A}{d}$$



4. Ans (2)

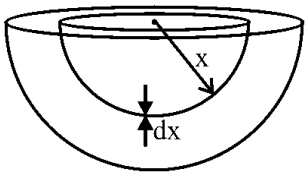
$N\sin\theta = mx\omega^2$

$N\cos\theta = mg$

$$\tan\theta = \frac{\omega^2}{g}x$$

$$dy = \frac{\omega^2}{g}xdx \Rightarrow y = \frac{\omega^2 x^2}{2g}$$

5. Ans (3)



$$V = \int_a^b \frac{K\left(2\pi x^2 dx\right)\rho}{x}$$

$$V = 2\pi k\rho\left[\frac{b^2 - a^2}{2}\right] = \pi\rho k\,(b^2 - a^2)$$

$b = 2a$

$v = 3\pi k\rho b^2$

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6. **Ans (2)**

For spherical surface

using $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$

$\Rightarrow \frac{n}{2R}-\frac{1}{\infty}=\frac{n-1}{R}$

$\Rightarrow n = 2n - 2$

$\Rightarrow n = 2$

7. **Ans (1)**

Time period is same for both tunnels i.e because its not depend on angle.

$T = 2\pi\sqrt{\frac{R}{g}}$

8. **Ans (9)**

$\vec{P}\perp\vec{Q} \Rightarrow \vec{P}\,.\,\vec{Q}- = 0$

$\therefore \quad (5\hat{i}+4\hat{j}+\alpha\hat{k})\cdot(\hat{i}+\hat{j}-\hat{k})=0$

or $5 + 4 - \alpha = 0$

$\Rightarrow \quad \alpha = 9$

9. **Ans (4)**

$-\mu mgx-\frac{1}{2}kx^2=0-\frac{1}{2}mv^2$

$v^2=\frac{1.44}{9}=\frac{4}{10} \Rightarrow N = 4$

10. **Ans (5)**

Displacement = $\left(3\hat{i}-7\hat{j}-9\hat{k}\right)$

$w = \vec{F}\,.\,\vec{x}$

$w = 9 - 49 + 45 = 5$ Joule

PART-2 : CHEMISTRY

SECTION-I

3. **Ans (C)**

$X \times 0.6 \times \frac{60}{100} = 200 \times 0.9 \times \frac{12}{100}$

$X = 60$ mL

4. **Ans (A)**

$W_{1NaoH}=\frac{30}{100}\times 300 = 90g$

$W_{2NaoH}=\frac{40}{100}\times 500 = 200$

$W_{t(NaoH)} = 200+90 = 290$

$W_{t(Soln)} = 300+500 = 800$

$D = 2\,\frac{800}{V}$

$V = 400$

$\frac{W}{V}=\frac{290}{400}\times 100$

$= 72.5$

7. **Ans (C)**

B removed $=\frac{1}{2}+\frac{1}{2}+4\times\frac{1}{8}=\frac{3}{2}$

A removed $= 1 + 2 \times \frac{1}{4}=\frac{3}{2}$

8. **Ans (A)**

B,D : ideal solution

C : Negative deviation

9. **Ans (A)**

$\frac{n}{N}=\frac{P^o-P}{P}$

10. **Ans (B)**

For ideal solution is $\Delta S_{mix} \neq 0$.

Due to mixing of gases randomness increases.

11. **Ans (C)**

Due to resonance, Bond order of O_3 is 1.5, so x will be between 121 pm and 148 pm.

12. **Ans (C)**

$d_{yz} + d_{yz}$ along $y -$ axis, forms $\pi-$bond.

$d_{yz} + d_{yz}$ along $z -$ axis, forms $\pi-$bond.

$d_{yz} + d_{yz}$ along $x -$ axis, forms $\delta-$bond.

$d_{xz} + d_{xz}$ along $x -$ axis, forms $\pi-$bond.

13. **Ans (A)**

According to Fajan's rule, polarisation of anion is influenced by charge and size of cation, more is the charge on cation, more is polarisation of anion.

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19. **Ans (D)**

$A_{(S)} \rightleftharpoons 2B_{(s)} = C_{(g)}$

$k = \frac{\ln}{t} \frac{P^{\infty} - P^{0}}{P^{\infty} - Pt}$

$k = \frac{\ln}{20} \frac{225 - 0}{225 - 150}$

$k = \frac{\ln}{20} \frac{225}{75}$

$k = 0.05 \ln 3 \text{ min}^{-1}$

PART-2 : CHEMISTRY

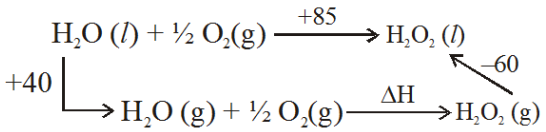
SECTION-II

1. **Ans (45)**

$H_2(g) + O_2(g) \longrightarrow H_2O_2(\ell)$; $\Delta H = -200$ .....(A)

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell)$; $\Delta H = -285$ .....(B)

Eq. (A) & Eq (B)



$85 = 40 + (2 \times \Delta H_{O-H} + \frac{1}{2} \times \Delta H_{O=O} - 2 \times \Delta H_{O-H} - \Delta H_{O=O}) - 60$

$\Delta H_{O-O} = 45$ kJ/mol

2. **Ans (5)**

$\frac{1}{4}P_4(s) + \frac{5}{2}Cl_2(g) \rightarrow PCl_5(s)$ ....(1)

$\frac{1}{2}P_4(s) + 3Cl_2 \longrightarrow 2PCl_3(\ell)$ ....(2)

$PCl_3(\ell) + Cl_2(g)\ PCl_5(s)$ ....(3)

On applying thermodynamic operation (1) = (3) $+ \frac{1}{2}$ (2)

$= -200 + \frac{1}{2}(-600)$

$= -500$ kJ $= -5 \times 10^2$ kJ

4. **Ans (4)**

ΔH solution is Lattice energy + Hydration energy.

$788 - 784 = 4$

7. **Ans (2)**

$w = -p_{ext}.\Delta V = -1\text{atm} \times (22 - 2)$ litre $= -20$ lit-atm $= -2000$ J $= -2$ kJ.

$|w| = 2$ kJ

9. **Ans (4)**

$\Delta H_f = 150 + 350 + 350 + q - 1000$

$-750 = -150 + q$

$q = -600$

$3X(g) \rightleftharpoons 3X^{\ominus}_{(g)} q = -600$

Electron Affinity

$X(g) \rightleftharpoons X^{\ominus}(g)\ q_1 = -\frac{600}{3} = -200$

So $\frac{q_1}{50} = -\frac{200}{50} = 4$

10. **Ans (2)**

O_2 : $(\sigma 1s^2)$ $(\sigma^* 1s^2)$ $(\sigma 2s^2)$ $(\sigma^* 2s^2)(\sigma 2p_z^2)$ $(\pi 2p_x^2 = \pi 2p_y^2)$ $(\pi^* 2p_x^1 = \pi^* 2p_y^2)$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (D)**

$\tan(180^\circ + \theta) = \tan\theta$

$\sum_{k=1^\circ}^{89^\circ} \log(\tan k)$

$= \log\{(\tan 1^\circ)(\tan 89^\circ)... (\tan 44^\circ).$

$(\tan 46^\circ).\ \tan 45^\circ\}$

$= \ln(1) = 0$

2. **Ans (D)**

$\because$ 'f' is odd f^{-1} is also odd

$\Rightarrow$ sum $= 0$

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3. Ans (C)

Clearly, let

$f'(x) = kx^2(x-2)(1+x^2),\ k \neq 0$

$= k(x^5 - 2x^4 + x^3 - 2x^2)$

$\Rightarrow f(x) = \frac{kx^3}{60}(10x^3 - 24x^2 + 15x - 40)$

$\therefore \frac{3f(2)}{f(1)} = \frac{3 \times 8(-26)}{-39} = 16$

4. Ans (C)

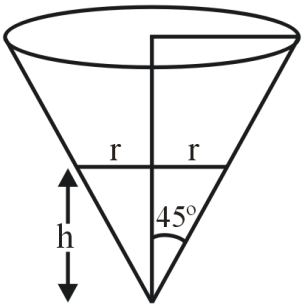
$t^2 + 3t - 8 = 2 \Rightarrow t = 2, -5$

$2t^2 - 2t - 5 = -1 \Rightarrow t = 2, -1$

So, at $t = 2$

$\frac{dy}{dx} = \left(\frac{\frac{dy}{dt}}{\frac{dx}{dt}}\right)_{t=2} = \frac{6}{7}$

5. Ans (B)



$\because \frac{h}{r} = \tan 45°$

$\frac{dv}{dt} = -2$

$\Rightarrow \frac{d}{dt}\left(\frac{1}{3}\pi r^2 h\right) = -2 \Rightarrow \frac{d}{dt}\left(\frac{1}{3}\pi r^3\right) = -2$

$\pi r^2 \frac{dr}{dt} = -2$ ...(i)

Also $\frac{d}{dt}(2\pi r) = 2\pi\frac{dr}{dt} = 2\pi\left(\frac{-2}{\pi r^2}\right) = \frac{-4}{r^2}$.

At $r = 2$, $\frac{d}{dt}(2\pi r) = \frac{-4}{4} = -1 \Rightarrow d = -1$

6. Ans (B)

$by^2 = (x+a)^3 \Rightarrow 2by.\frac{dy}{dx} = 3(x+a)^2 \Rightarrow \frac{dy}{dx} = \frac{3}{2by}(x+a)^2$

$\therefore$ Subnormal $= y\frac{dy}{dx} = \frac{3}{2b}(x+a)^2$

$\therefore$ Subtangent $= \frac{y}{\left(\frac{dy}{dx}\right)} = \frac{y}{\frac{3(x+a)^2}{2by}} = \frac{2by^2}{3(x+a)^2}$

$= \frac{2b\frac{(x+a)^3}{b}}{3(x+a)^2} = \frac{2}{3}(x+a)$

$\therefore$ (Subtangent)$^2 = \frac{4}{9}(x+a)^2$

and $\frac{(\text{Subtangent})^2}{\text{subnormal}} = \frac{\frac{4}{9}(x+a)^2}{\frac{3}{2b}(x+a)^2} = \frac{8b}{27}$

$\Rightarrow$ (Subtangent)^2 = constant × (subnormal)

$\therefore$ (Subtangent)$^2 \propto$ (Subnormal).

7. Ans (A)

Let the roots be $A-3d, A-d, A+d, A+3d$

sum of roots $= 0 \Rightarrow A = 0$

P.O.R $= 9d^4 = 16K^2 \Rightarrow d^2 = \frac{4K}{3}$ ...(1)

Sum of roots taken two at a time i.e.

$\sum x_1x_2 = -10d^2 = -(12K+5)$

$10\left(\frac{4K}{3}\right) = 12K + 5 \Rightarrow K = \frac{15}{4} = \frac{a}{b}$

$\therefore (a - 3b) = 3$

8. Ans (C)

$\left(x^4 + \frac{1}{x^4} + 1\right)\left(y^4 + \frac{1}{y^4} - 1\right)\left(z^4 + \frac{1}{z^4} + \frac{1}{3}\right)$

$\because x^4 + \frac{1}{x^4} \geqslant 2$

$\Rightarrow 3 \times (1) \times \left(2 + \frac{1}{3}\right) = 7$

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9. Ans (B)

$\Rightarrow S_{n+2}(S_n - S_{n-1}) - S_{n+1}(S_n - S_{n-1})$

$\Rightarrow S_{n+2}\, T_n - S_{n+1}T_n$

$\Rightarrow T_n(S_{n+2} - S_{n+1})$

$\Rightarrow T_n(T_{n+2})$

$\Rightarrow \dfrac{21}{(6n-5)(6n+7)}$

$\text{sum} \Rightarrow \sum_{n=1}^{\infty} \dfrac{7}{4}\left[\dfrac{1}{6n-5} - \dfrac{1}{6n+7}\right]$

$\therefore$ sum = 2

10. Ans (B)

Let $f(x) = 2x^3 - 3x^2 + p$

$f'(x) = 6x(x-1)$

$f(0).f(1) \le 0$ (Not necesserily distrinct)

$p \in [0, 1]$

11. Ans (B)

$t_r = \dfrac{\sin(2^r\theta)\sin^2(2^{r-1}\theta)}{2^r}$; $r = 0, 1, 2, ..., n$

$\Rightarrow t_r = \dfrac{\sin(2^r\theta)\,[1 - \cos(2^r\theta)]}{2^{(r+1)}}$

$= \dfrac{2\sin(2^r\theta) - \sin\left(2^{(r+1)}\theta\right)}{2^{(r+2)}}$

$\Rightarrow t_r = \dfrac{\sin(2^r\theta)}{2^{(r+1)}} - \dfrac{\sin\left(2^{(r+1)}\theta\right)}{2^{(r+2)}}$

$\Rightarrow f(n,\theta) = \dfrac{\sin\theta}{2} - \dfrac{\sin\left(2^{(n+1)}\theta\right)}{2^{(n+2)}}$

12. Ans (A)

$\dfrac{\cos 20^\circ}{\sin 50^\circ \sin 110^\circ} + \dfrac{\cos 50^\circ}{\sin 110^\circ . \sin 20^\circ} + \dfrac{\cos 110^\circ}{\sin 20^\circ \sin 50^\circ}$

$= \dfrac{1}{2}\left[\dfrac{\sin 40^\circ + \sin 100^\circ + \sin 220^\circ}{\sin 20^\circ \sin 50^\circ \sin 110^\circ}\right]$

$= \dfrac{1}{2}\left[\dfrac{4\sin 20^\circ \sin 50^\circ \sin 110^\circ}{\sin 20^\circ \sin 50^\circ \sin 110^\circ}\right] = 2$

13. Ans (C)

$x \ge 0$

$\cos^{-1}x + \cos^{-1}2x = \pi$

$2x = \cos(\pi - \cos^{-1}x)$

$= -\cos(\cos^{-1}x)$

$= -x \quad \Rightarrow \quad x = 0$

$\underline{x < 0}$

$\cos^{-1}(-x) + \cos^{-1}(-2x) = \pi$

$\pi - \cos^{-1}x + \pi - \cos^{-1}(2x) = \pi$

$\cos^{-1}x + \cos^{-1}(2x) = \pi$

14. Ans (C)

Let $n = \dfrac{1}{x}$

$\ell = \lim_{x\to 0} \dfrac{e(1-x)^{1/x} - 1}{(1/x)^{\alpha}} = \lim_{x\to 0} \dfrac{e \cdot e^{\frac{\ell n(1-x)}{x}} - 1}{x^{-\alpha}}$

$= \lim_{x\to 0}\left(\dfrac{\frac{\ell n(1-x)}{x} + 1}{x^{-\alpha}}\right) = \lim_{x\to 0}\left(\dfrac{-\frac{x}{2} - \frac{x^2}{3}.....}{x^{-\alpha}}\right)$

for limit to exist $\alpha = -1$

$\ell = -\dfrac{1}{2}$

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15. Ans (A)

Given $\cot\frac{A}{2}\cdot\cot\frac{C}{2}=3$

$$\Rightarrow \frac{\cos\frac{A}{2}\cdot\cos\frac{C}{2}}{\sin\frac{A}{2}\cdot\sin\frac{C}{2}}=3 \Rightarrow \frac{\cos\frac{A-C}{2}}{\cos\frac{A+C}{2}}=2$$

(using componendo and dividendo)

$$\Rightarrow \frac{2\sin\frac{A+C}{2}\cos\frac{A-C}{2}}{2\sin\frac{A+C}{2}\cdot\cos\frac{A+C}{2}}=2$$

$= 2\sin B = \sin A + \sin C$

16. Ans (B)

$$\lim_{x\to0}\frac{f'(x)+2^2f'(2x)+3^2f'(3x)}{2x}$$

$$\lim_{x\to0}\frac{f''(x)+8f''(2x)+27f''(3x)}{2}=36$$

17. Ans (D)

Putting x = 0 in the given equation,

We have f(0) = f′′′(3) and putting x = 1, we get

f(1) = 1 + f′(1) + f′′(2) + f′′′(3).

Thus, f(1) − f(0) = 1 + f′(1) + f′′(2).

Also differentiating the given equation, we have

$f'(x) = 3x^2 + 2x\,f'(1) + f''(2)$ .... (i)

f′′(x) = 6x + 2f′(1), f′′′(x) = 6

Thus f′′′(3) = 6 and f′′(2) = 12 + 2f′(1)

Putting x = 1 in (i), we have

f′(1) = 3 + 2f′(1) + f′′(2) = 3 + 2 f′(1) + 12 + 2 f′(1)

= 15 + 4f′(1)

f′(1) = −5 and so f′′(2) = 12 − 10 = 2.

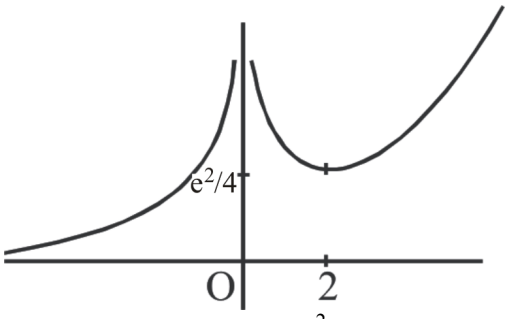
⇒ f(2) = −2

18. Ans (B)

Let $f(x) = e^x x^{-2}$

$$f'(x) = \frac{e^x}{x^3}(x-2)$$

$$f(2) = \frac{e^2}{4}$$



Clearly from graph, $k > \frac{e^2}{4}$

$\Rightarrow k_{min} = 2.$

19. Ans (A)

$$x^2 + \left(y-\frac{5}{3}\right)^2 = 1$$

$y^2 = x^3$, $P(t^2,t^3)$

Normal at point P is

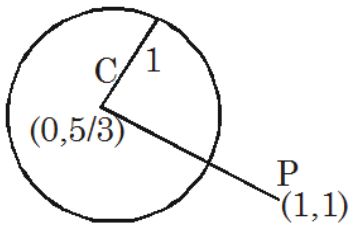
$$y - t^3 = -\frac{2}{3t}(x-t^2)$$

it passes through $\left(0,\frac{5}{3}\right)$

$3t^3 + 2t - 5 = 0$

$(t-1)(3t^2+3t+5) = 0$

D < 0



shortest distance = CP − 1

$$= \frac{\sqrt{13}}{3} - 1$$

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20. Ans (D)

For $f(0^-)$, let sinx = h

$$\Rightarrow f(0^-) = \lim_{h\to 0} \frac{h(-1)^2 \ell n2}{\ell n(1+h)} = \ell n2$$

for $f(0^+)$ let tanx = h

$$\Rightarrow f(0^+) = \lim_{h\to 0} \frac{\ell n(e^{h^2} + 2\sqrt{h})}{\tan\sqrt{h}}$$

$$\lim_{h\to 0} \frac{e^{h^2} - 1 + 2\sqrt{h}}{\tan\sqrt{h}} \Rightarrow f(0^+) = 2$$

PART-3 : MATHEMATICS

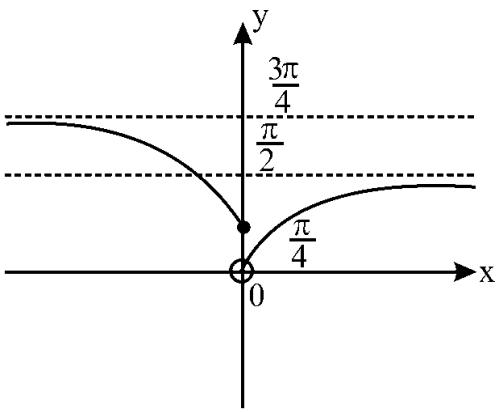
SECTION-II

1. Ans (1)

$$\text{A.M.} = \frac{\log_a(c-b) + \log_a(c+b)}{2} = 1$$

2. Ans (5)

For $x \leqslant 0$, $f(x) = \cos^{-1}\left(\frac{1+x}{\sqrt{2(1+x^2)}}\right)$



Put $x = \tan\theta \Rightarrow \theta = \tan^{-1}x$; $\theta \in \left(\frac{-\pi}{2}, 0\right]$

$$= \cos^{-1}\cos\left(\theta - \frac{\pi}{4}\right) = \frac{\pi}{4} - \tan^{-1}x.$$

3. Ans (1)

$$\frac{\alpha}{\alpha-1} + \frac{\alpha+1}{\alpha} = -\frac{b}{a} \quad \text{....(i)}$$

$$\& \quad \frac{\alpha}{\alpha-1}\frac{\alpha+1}{\alpha} = \frac{c}{a} \Rightarrow \alpha = \frac{c+a}{c-a}$$

Put the value of α in equation (i) & simplify

$$\Rightarrow (c+a)^2 + 4ac = -2b(c+a)$$

$$(c+a)^2 + 2b(c+a) + b^2 = b^2 - 4ac$$

$$(a+b+c)^2 = b^2 - 4ac$$

$$\Rightarrow \frac{(a+b+c)^2 + 4ac}{b^2} = 1$$

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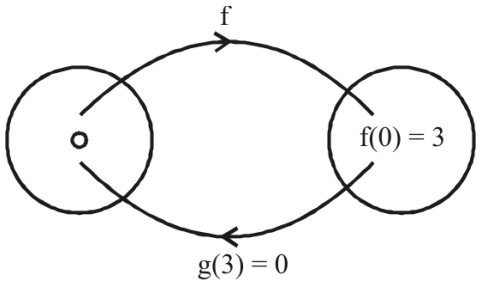
4. **Ans (2)**

$f(x) = x + \cos x + 2$

$f'(x) = 1 - \sin x$ and $f'(0) = 1$

$f''(x) = -\cos x$ and $f''(0) = -1$

Now, $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} \Rightarrow g'(y) = \frac{1}{f'(x)}$



When $y = 3$ then $x = 0$

$\therefore \quad g'(3) = \frac{1}{f'(0)} = 1$ .....(1)

Again $\frac{d^2x}{dy^2} = \frac{d}{dy}\left(\frac{1}{\frac{dy}{dx}}\right)$

$= \frac{d}{dx}\left(\frac{1}{\frac{dy}{dx}}\right) \cdot \frac{dx}{dy} = \frac{-1}{\left(\frac{dy}{dx}\right)^3} \cdot \frac{d^2y}{dx^2}$

$\Rightarrow \quad g''(y) = \frac{-f(x)}{(f(x))^2}$ or

$g''(3) = \frac{-f(0)}{(f(0))^3} = \frac{1}{1}$

$g''(3) = 1$ .....(2)

Hence, $g'(3) + g''(3) = 1 + 1 = 2$. Ans.]

Alternatively: Clearly, f is a bijective mapping.

Also, $f(0) = 3$

$\therefore \quad g'(3) = \frac{1}{f'(0)}$ .....(1)

and $g''(3) = \frac{-f(0)}{(f'(0))^3}$ .....(2)

As, $f(x) = x + \cos x + 2$

$\Rightarrow f'(x) = 1 - \sin x \Rightarrow f'(0) = 1$

and $f''(x) = -\cos x \Rightarrow f'(1) = -1$

$\therefore$ From (1) and (2), we get

$g'(3) = \frac{1}{1} = 1$ and $g''(3) = \frac{-(-1)}{(1)^3} = 1.$

Hence, $\left(g'(3) + g(3)\right) = 1 + 1 = 2$. Ans.

5. **Ans (1)**

$$f\left(\frac{2}{x-2}\right) = \frac{1}{\frac{4}{(x-2)^2} - \frac{34}{(x-2)} + 66}$$

$$= \frac{(x-2)^2}{4 - 34(x-2) + 66(x-2)^2}$$

$$= \frac{(x-2)^2}{2(3x-7)(11x-24)}$$

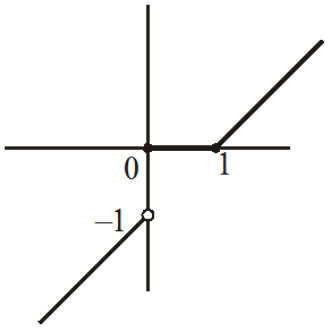
$\Rightarrow f\left(\frac{2}{x-2}\right)$ is discontinuous at $x = 2, \frac{7}{3}, \frac{24}{11}$

But removable discontinuity at $x = 2$ only.

6. **Ans (8)**

$f(x) + f(1 - x) = 1$

7. **Ans (2)**



Not differentiable at $x = 0, 1$

8. **Ans (2)**

$f(x) + f(y) = f(x)f(y) + f(xy)$

partial differentiation wrt x

$f'(x) = f'(x)f(y) + f'(xy).(y)$

put $x = 1$

$-2 = -2f(y) + f'(y)y$

get $f(y) = 1 - y^2$

$\Rightarrow f(x) = 1 - x^2$

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9. **Ans (6)**

$f(1) = 3 \Rightarrow \beta - \alpha = 3$

$f'(1) = -1 \Rightarrow 2\alpha = 3$

$\Rightarrow \alpha = \frac{3}{2}$ & $\beta = \frac{9}{2}$

10. **Ans (1)**

$$y = 3\tan A \tan\left(\frac{\pi}{3} - A\right)$$

$$y = 3\tan A\left(\frac{\sqrt{3} - \tan A}{1 + \sqrt{3}\tan A}\right)$$

$$y = 3\frac{\left(\sqrt{3}t - t^2\right)}{1 + \sqrt{3}t}, t > 0$$

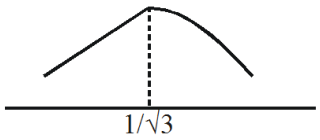
$$y' = 3\left[\frac{\left(1 + \sqrt{3}t\right)\left(\sqrt{3} - 2t\right) - \left(\sqrt{3}t - t^2\right)\left(+\sqrt{3}\right)}{\left(1 - \sqrt{3}t\right)^2}\right]$$

$$y' = \frac{3\left(t + \sqrt{3}\right)\left(-\sqrt{3}t + 1\right)}{\left(1 - \sqrt{3}t\right)^2}$$

$y' > 0 \Rightarrow \left(t + \sqrt{3}\right)\left(\sqrt{3}t - 1\right) < 0$

$$-\sqrt{3} < t < \frac{1}{\sqrt{3}}$$

$y' < 0 \Rightarrow t < -\sqrt{3}, t > \frac{1}{\sqrt{3}}$



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